

PIZZA SALES ANALYTICS & BUSINESS INTELLIGENCE

AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

BharatCares in association with **AICTE**

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1. Abstract

This project presents a data analytics and business intelligence solution for pizza sales data. The solution uses interactive Power BI dashboards to monitor key performance indicators, analyze revenue and order trends, compare product/category/size performance, examine time and operational patterns, and explore business drivers through advanced Power BI analytics. The project is designed to convert transactional data into information, insights and potential business actions.

2. Introduction

Sales transactions contain useful information about revenue, orders, product demand and operating patterns. A business intelligence dashboard can organize these measures into a form that helps users monitor performance and investigate areas that require attention. This project develops such an analytical view for pizza sales.

3. Problem Statement

Pizza sales data can become difficult to interpret when revenue, product, category, size and time information are viewed separately. The project addresses this by bringing the major analytical views together in an interactive dashboard and supporting the dashboard with a reproducible Python analysis workflow.

4. Objectives

- Analyze overall revenue, orders and quantity sold.
- Monitor year-over-year revenue performance.
- Identify high- and low-performing pizza products.
- Analyze performance by pizza category and size.
- Analyze revenue by day, hour, day-part and month.
- Provide interactive filtering and detailed product analysis.
- Use advanced Power BI analytics features to explore business drivers.
- Translate supported findings into business-oriented recommendations.

5. Technologies Used

- Microsoft Power BI
- Power Query
- DAX
- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

- GitHub

6. Project Methodology

The project follows the analytical flow: Raw Sales Data → Data Preparation → Exploratory Analysis → KPI Analysis → Power BI Visualization → Advanced Analytics → Business Insights → Recommendations.

7. Dashboard Structure

Dashboard Page	Purpose
Executive Dashboard	KPI monitoring, revenue comparison, category and size analysis.
Product Performance	Product ranking and analysis of products that may be promoted or reviewed.
Time & Operational Analysis	Revenue analysis by day, hour, day-part and month.
Advanced Analytics	Decomposition Tree, Key Influencers, Q&A and ranking analysis.
Pizza Details	Detailed product-level analysis.
Pizza Tooltip	Contextual product information including revenue, orders, quantity, contribution and rank.

8. Key Performance Indicators

- Total Revenue
- Total Orders
- Total Pizzas Sold
- Average Order Value
- Revenue YoY %
- Average Pizzas per Order
- Revenue % of Total
- Pizza/Product Ranking

9. Analytics Performed

9.1 Executive Analysis

The Executive Dashboard provides a consolidated view of major KPIs and allows users to filter the analysis by pizza category, pizza size and day part.

9.2 Product Analysis

The Product Performance page provides product ranking and supports identification of products that could be considered for promotion or further review.

9.3 Time and Operational Analysis

Revenue is examined across days, hours, day parts and months to identify temporal patterns that can support operational planning.

9.4 Advanced Analytics

The Advanced Analytics page uses Power BI Decomposition Tree, Key Influencers, Q&A and ranking analysis to explore relationships and potential drivers in the sales data.

10. Business Insight Framework

The final analysis should express findings using the structure: Finding → Evidence → Business Implication → Recommended Action. Specific numerical claims should be inserted only after validating the dataset and dashboard outputs.

11. Business Recommendations

- Monitor high-level revenue and order KPIs regularly.
- Use product performance analysis to identify products suitable for promotion or review.
- Use peak day/hour/day-part patterns for operational and inventory planning.
- Monitor year-over-year changes to identify periods requiring further investigation.
- Use advanced analytics to investigate possible drivers before taking business action.

12. Dataset

The final submission must contain the link to the independently verified dataset used by the project. The internship instructions state that the exact dataset used in the internship/masterclasses must not be reused.

Dataset : Pizza Sales Dataset

Important: Replace the placeholder above before submission. If the dataset license does not permit redistribution, provide the public source link and instructions rather than uploading the raw dataset.

13. Python Notebook

The accompanying Jupyter Notebook performs dataset inspection, data-quality checks, KPI calculations, category/product analysis and time analysis. The notebook expects the independently verified CSV dataset to be placed in the data folder.

14. Project Files

Required File	Purpose
PushpamSharma_PizzaSalesAnalytics.ipynb	Complete Python/Jupyter project code.
requirements.txt	Python libraries/dependencies required to run the notebook.

PushpamSharma_ProjectReport.docx	Complete project documentation.
README.md	Project overview, dataset link, technologies and setup/run information.

15. Setup and Execution

Create a virtual environment in Windows PowerShell:

```
python -m venv .venv
```

Activate it:

```
.\.venv\Scripts\Activate.ps1
```

Install dependencies:

```
pip install -r requirements.txt
```

Place the verified CSV dataset in the data folder, then launch Jupyter:

```
jupyter notebook
```

16. Limitations

The Power BI report and the Python workflow are separate project layers. The PBIX file does not provide a reliable standalone Python CSV for automatic extraction in this submission workflow. Therefore, the Python notebook is designed to use the verified source dataset supplied by the student. Dataset-specific numerical conclusions must be validated before final submission.

17. Conclusion

The project provides an interactive business intelligence solution for pizza sales analysis. It combines KPI monitoring, product analysis, time analysis and advanced Power BI analytics. Its objective is to support a structured transition from data to information, insights and business-oriented action.